

Type KR··LL
(with cage, sealed type)

Accessories

Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
30～ 35	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5～1M16×1.5
40～180	JIS A-PT1/8	PT1/8×7ℓ	1M18×1.5～1M64×3

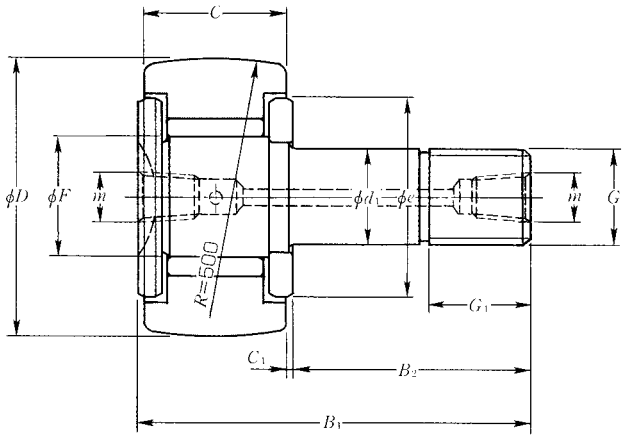
※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Load ratings of truck N		Load ratings of truck kgf		Limiting speeds r/min		Maximum tightening torque		Bearing numbers				Mass	Stud diameter
								Without seal		With seal			
Spherical outer ring	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	grease	oil	N·m	kgf·m	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	kg (approx.)	mm
1 080	3 400	110	350	※19 000	※25 000	3	0.3	**KRT16**	**KRT16X**	**KRT16LL**	**KRT16XLL**	0.019	**6**
1 380	4 050	141	415	※15 000	※20 000	8	0.8	**KRT19**	**KRT19X**	**KRT19LL**	**KRT19XLL**	0.031	**8**
1 690	5 150	172	525	※12 000	※16 000	14	1.4	**KRT22**	**KRT22X**	**KRT22LL**	**KRT22XLL**	0.046	**10**
2 120	6 100	216	620	※12 000	※16 000	14	1.4	**KRT26**	**KRT26X**	**KRT26LL**	**KRT26XLL**	0.059	**10**
2 620	7 700	267	785	10 000	※13 000	20	2	**KRT30**	**KRT30X**	**KRT30LL**	**KRT30XLL**	0.087	**12**
2 860	8 200	291	835	10 000	※13 000	20	2	**KRT32**	**KRT32X**	**KRT32LL**	**KRT32XLL**	0.097	**12**
3 200	11 900	325	1 220	8 000	※11 000	52	5.3	**KRT35**	**KRT35X**	**KRT35LL**	**KRT35XLL**	0.169	**16**
3 850	14 500	390	1 480	7 000	9 000	76	7.8	**KRT40**	**KRT40X**	**KRT40LL**	**KRT40XLL**	0.248	**18**
4 700	21 000	480	2 150	6 000	8 000	98	10	**KRT47**	**KRT47X**	**KRT47LL**	**KRT47XLL**	0.386	**20**
5 550	23 300	565	2 370	6 000	8 000	98	10	**KRT52**	**KRT52X**	**KRT52LL**	**KRT52XLL**	0.461	**20**
6 950	34 500	710	3 500	5 000	6 500	178	18	**KRT62**	**KRT62X**	**KRT62LL**	**KRT62XLL**	0.790	**24**
8 050	38 500	820	3 900	5 000	6 500	178	18	**KRT72**	**KRT72X**	**KRT72LL**	**KRT72XLL**	1.04	**24**
9 800	53 000	1 000	5 400	4 000	5 500	360	37	**KRT80**	**KRT80X**	**KRT80LL**	**KRT80XLL**	1.55	**30**
10 400	56 000	1 060	5 750	4 000	5 500	360	37	**KRT85**	**KRT85X**	**KRT85LL**	**KRT85XLL**	1.74	**30**
11 400	59 000	1 160	6 100	4 000	5 500	360	37	**KRT90**	**KRT90X**	**KRT90LL**	**KRT90XLL**	1.95	**30**

Remarks: The allowable running speed of **Type KRT··LL** and **KRT··XLL** with ※marked seal is approximately 10 000r/min.

Metric series

Type KRT
Type RT··X
Type KRT··LL
Type KRT··XLL

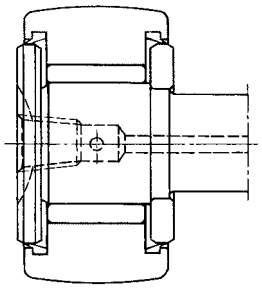


Type KRV
(Full-complement roller type)

D 16～90mm

Boundary dimensions mm											Basic load ratings dynamic N	static N	dynamic kgf	static kgf
D 1)	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	C_r	C_{or}	C_r	C_{or}
16	11	$6\ ^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	12	M4×0.7 2)	6 500	9 350	665	955
19	11	$8\ ^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	14	M4×0.7 2)	7 450	11 700	760	1 190
22	12	$10\ ^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	8 200	14 000	840	1 420
26	12	$10\ ^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	8 200	14 000	840	1 420
30	14	$12\ ^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	12 000	20 300	1 230	2 070
32	14	$12\ ^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	12 000	20 300	1 230	2 070
35	18	$16\ ^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	27	PT1/8	17 600	34 000	1 790	3 500
40	20	$18\ ^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	32	PT1/8	19 400	42 000	1 980	4 250
47	24	$20\ ^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	28 800	61 000	2 940	6 250
52	24	$20\ ^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	28 800	61 000	2 940	6 250
62	29	$24\ ^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	39 500	98 500	4 000	10 000
72	29	$24\ ^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	39 500	98 500	4 000	10 000
80	35	$30\ ^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	58 000	147 000	5 900	15 000
90	35	$30\ ^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	58 000	147 000	5 900	15 000

Note 1) The dimensional tolerances for the outer ring outer diameter *D* of **Type KRVT··X** and **KRVT··XLL** with cylindrical outer surface conform to JIS Class-0.
2) A tapped hole is provided at front side only (on left end-face in the above sketch).



Type KRVT··LL
(Full-complement roller and sealed type)

Accessories

Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
30～ 35	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5～1M16×1.5
40～180	JIS A-PT1/8	PT1/8×7 ℓ	1M18×1.5～1M64×3

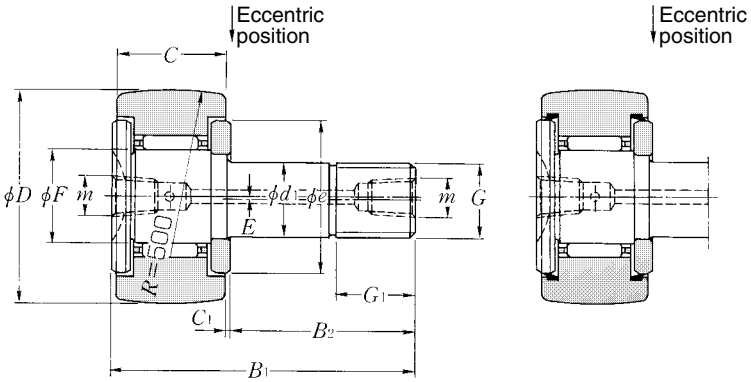
※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Load ratings of truck N		Load ratings of truck kgf		Limiting speeds r/min		Maximum tightening torque		Bearing numbers				Mass	Stud diameter
								Without seal		With seal			
Spherical outer ring	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	grease	oil	N·m	kgf·m	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	kg (approx.)	mm
1 080	3 400	110	350	※13 000	※16 000	3	0.3	**KRVT16**	**KRVT16X**	**KRVT16LL**	**KRVT16XLL**	0.020	**6**
1 380	4 050	141	415	10 000	※13 000	8	0.8	**KRVT19**	**KRVT19X**	**KRVT19LL**	**KRVT19XLL**	0.032	**8**
1 690	5 150	172	525	8 500	※11 000	14	1.4	**KRVT22**	**KRVT22X**	**KRVT22LL**	**KRVT22XLL**	0.047	**10**
2 120	6 100	216	620	8 500	※11 000	14	1.4	**KRVT26**	**KRVT26X**	**KRVT26LL**	**KRVT26XLL**	0.061	**10**
2 620	7 700	267	785	6 500	8 500	20	2	**KRVT30**	**KRVT30X**	**KRVT30LL**	**KRVT30XLL**	0.089	**12**
2 860	8 200	291	835	6 500	8 500	20	2	**KRVT32**	**KRVT32X**	**KRVT32LL**	**KRVT32XLL**	0.100	**12**
3 200	11 900	325	1 220	5 500	7 000	52	5.3	**KRVT35**	**KRVT35X**	**KRVT35LL**	**KRVT35XLL**	0.172	**16**
3 850	14 500	390	1 480	4 500	6 000	76	7.8	**KRVT40**	**KRVT40X**	**KRVT40LL**	**KRVT40XLL**	0.252	**18**
4 700	21 000	480	2 150	4 000	5 000	98	10	**KRVT47**	**KRVT47X**	**KRVT47LL**	**KRVT47XLL**	0.390	**20**
5 550	23 300	565	2 370	4 000	5 000	98	10	**KRVT52**	**KRVT52X**	**KRVT52LL**	**KRVT52XLL**	0.465	**20**
6 950	34 500	710	3 500	3 300	4 500	178	18	**KRVT62**	**KRVT62X**	**KRVT62LL**	**KRVT62XLL**	0.800	**24**
8 050	38 500	820	3 900	3 300	4 500	178	18	**KRVT72**	**KRVT72X**	**KRVT72LL**	**KRVT72XLL**	1.05	**24**
9 800	53 000	1 000	5 400	2 600	3 500	360	37	**KRVT80**	**KRVT80X**	**KRVT80LL**	**KRVT80XLL**	1.56	**30**
11 400	59 000	1 160	6 100	2 600	3 500	360	37	**KRVT90**	**KRVT90X**	**KRVT90LL**	**KRVT90XLL**	1.97	**30**

Remarks: The allowable running speed of **Type KRVT··LL** and **KRVT··XLL** with ※marked seal is approximately 10 000r/min.

Metric Series

Type KRU
Type KRU··LL
Type KRVU
Type KRVU··LL



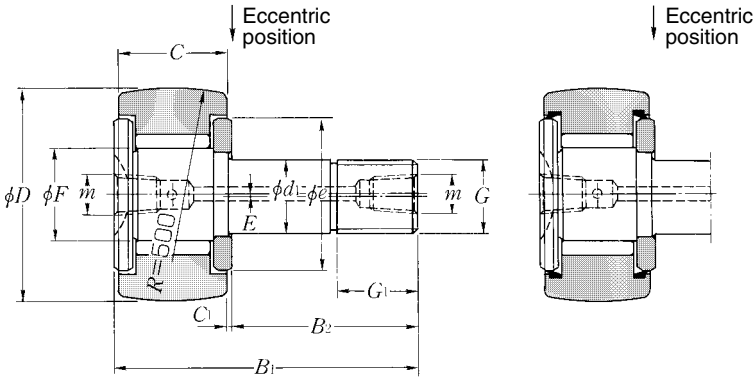
Type KRU
(With cage)

Type KRU··LL
(with cage, sealed type)

D 16～80mm

Boundary dimensions mm												Basic load ratings dynamic N	static N	dynamic kgf	static kgf
D $^{0}_{-0.050}$	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	E eccentricity	C_r	C_{or}	C_r	C_{or}
16	11	6 $^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	12	M4×0.7[2)]	0.25	4 050	4 200	415	430
	11	6 $^{0}_{-0.012}$	8	28	16	M 6×1	8	0.6	12	M4×0.7[2)]	0.25	6 500	9 350	665	955
19	11	8 $^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	14	M4×0.7[2)]	0.25	4 750	5 400	480	555
	11	8 $^{0}_{-0.015}$	10	32	20	M 8×1.25	10	0.6	14	M4×0.7[2)]	0.25	7 450	11 700	760	1 190
22	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	0.3	5 300	6 650	540	680
	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	0.3	8 200	14 000	840	1 420
26	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	0.3	5 300	6 650	540	680
	12	10 $^{0}_{-0.015}$	12	36	23	M10×1.25	12	0.6	17	M4×0.7	0.3	8 200	14 000	840	1 420
30	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	0.4	7 850	9 650	800	985
	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	0.4	12 000	20 300	1 230	2 070
32	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	0.4	7 850	9 650	800	985
	14	12 $^{0}_{-0.018}$	15	40	25	M12×1.5	13	0.6	23	M6×0.75	0.4	12 000	20 300	1 230	2 070
35	18	16 $^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	27	M6×0.75	0.5	12 500	18 900	1 280	1 930
	18	16 $^{0}_{-0.018}$	18	52	32.5	M16×1.5	17	0.8	27	M6×0.75	0.5	18 000	36 500	1 840	3 700
40	20	18 $^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	32	PT1/8	0.6	14 000	22 800	1 430	2 330
	20	18 $^{0}_{-0.018}$	22	58	36.5	M18×1.5	19	0.8	32	PT1/8	0.6	19 400	42 000	1 980	4 250
47	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	0.7	20 700	33 500	2 110	3 450
	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	0.7	28 800	61 000	2 940	6 250
52	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	0.7	20 700	33 500	2 110	3 450
	24	20 $^{0}_{-0.021}$	25	66	40.5	M20×1.5	21	0.8	37	PT1/8	0.7	28 800	61 000	2 940	6 250
62	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	0.8	28 900	55 000	2 950	5 600
	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	0.8	39 500	98 500	4 000	10 000
72	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	0.8	28 900	55 000	2 950	5 600
	29	24 $^{0}_{-0.021}$	30	80	49.5	M24×1.5	25	0.8	44	PT1/8	0.8	39 500	98 500	4 000	10 000
80	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	1.0	45 000	88 500	4 600	9 050
	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	1.0	58 000	147 000	5 900	15 000

Note 1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter D which conforms to JIS Class-0. **Ex. KRU52XLL**
2) A tapped hole is provided at front side only (on left side-face in the above sketch).



Type KRVU
(Full complement)

Type KRVU··LL
(Full complement sealed type)

Accessories

Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
16～26	NIP-X30	M4×0.7 ×4ℓ	1M 6×1 ～1M10×1.25
30～35	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5～1M16×1.5
40～90	JIS A-PT1/8	PT1/8×7 ℓ	1M18×1.5～1M30×1.5

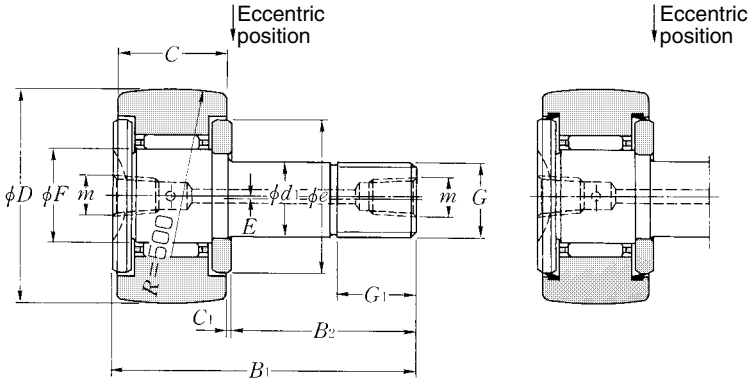
※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Load ratings of truck N		Load ratings of truck kgf		Limiting speeds r/min		Maximum tightening torque		Bearing numbers		Mass (approx.) kg	
Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	grease	oil	N·m	kgf·m	Type KRU [1] Type KRVU	Type KRU··ŁL [1] Type KRVU··ŁL	Type KRU	Type KRVU
1 080	3 400	110	350	※19 000	※25 000	3	0.3	**KRU16**	**KRU16LL**	0.019	—
1 080	3 400	110	350	※13 000	※16 000	3	0.3	**KRVU16**	**KRVU16LL**	—	0.020
1 380	4 050	141	415	※15 000	※20 000	8	0.8	**KRU19**	**KRU19LL**	0.031	—
1 380	4 050	141	415	10 000	※13 000	8	0.8	**KRVU19**	**KRVU19LL**	—	0.032
1 690	5 150	172	525	※12 000	※16 000	14	1.4	**KRU22**	**KRU22LL**	0.046	—
1 690	5 150	172	525	8 500	※11 000	14	1.4	**KRVU22**	**KRVU22LL**	—	0.047
2 120	6 100	216	620	※12 000	※16 000	14	1.4	**KRU26**	**KRU26LL**	0.059	—
2 120	6 100	216	620	8 500	※11 000	14	1.4	**KRVU26**	**KRVU26LL**	—	0.061
2 620	7 700	267	785	10 000	※13 000	20	2	**KRU30**	**KRU30LL**	0.087	—
2 620	7 700	267	785	6 500	8 500	20	2	**KRVU30**	**KRVU30LL**	—	0.089
2 860	8 200	291	835	10 000	※13 000	20	2	**KRU32**	**KRU32LL**	0.097	—
2 860	8 200	291	835	6 500	8 500	20	2	**KRVU32**	**KRVU32LL**	—	0.100
3 200	11 900	325	1 220	8 000	※11 000	52	5.3	**KRU35**	**KRU35LL**	0.169	—
3 200	11 900	325	1 220	5 500	7 000	52	5.3	**KRVU35**	**KRVU35LL**	—	0.172
3 850	14 500	390	1 480	7 000	9 000	76	7.8	**KRU40**	**KRU40LL**	0.248	—
3 850	14 500	390	1 480	4 500	6 000	76	7.8	**KRVU40**	**KRVU40LL**	—	0.252
4 700	21 000	480	2 150	6 000	8 000	98	10	**KRU47**	**KRU47LL**	0.386	—
4 700	21 000	480	2 150	4 000	5 000	98	10	**KRVU47**	**KRVU47LL**	—	0.390
5 550	23 300	565	2 370	6 000	8 000	98	10	**KRU52**	**KRU52LL**	0.461	—
5 550	23 300	565	2 370	4 000	5 000	98	10	**KRVU52**	**KRVU52LL**	—	0.465
6 950	34 500	710	3 500	5 000	6 500	178	18	**KRU62**	**KRU62LL**	0.790	—
6 950	34 500	710	3 500	3 300	4 500	178	18	**KRVU62**	**KRVU62LL**	—	0.800
8 050	38 500	820	3 900	5 000	6 500	178	18	**KRU72**	**KRU72LL**	1.04	—
8 050	38 500	820	3 900	3 300	4 500	178	18	**KRVU72**	**KRVU72LL**	—	1.05
9 800	53 000	1 000	5 400	4 000	5 500	360	37	**KRU80**	**KRU80LL**	1.55	—
9 800	53 000	1 000	5 400	2 600	3 500	360	37	**KRVU80**	**KRVU80LL**	—	1.56

Remarks: For the cam follower with ※marked seal, the allowable running speed is approximately 10 000 r/min.

Metric Series

Type KRU
Type KRU··LL
Type KRVU
Type KRVU··LL



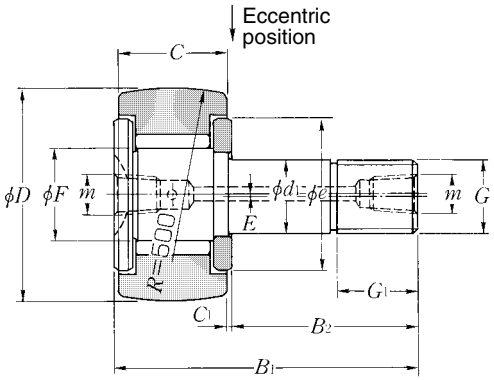
Type KRU (With cage)

Type KRU··LL (with cage, sealed type)

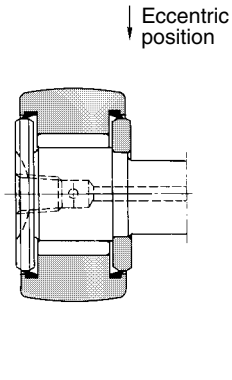
D 85～90mm

Boundary dimensions mm												Basic load ratings dynamic N	static N	dynamic kgf	static kgf
D $^{0}_{-0.050}$	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	E eccentricity	C_r	C_{or}	C_r	C_{or}
85	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	1.0	45 000	88 500	4 600	9 050
90	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	1.0	45 000	88 500	4 600	9 050
	35	30 $^{0}_{-0.021}$	38	100	63	M30×1.5	32	1	53	PT1/8	1.0	58 000	147 000	5 900	15 000

Note 1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter D which conforms to JIS Class-0. **Ex. KRU85XLL**



Type KRVU
(Full complement)



Type KRVU··LL
(Full complement sealed type)

Accessories

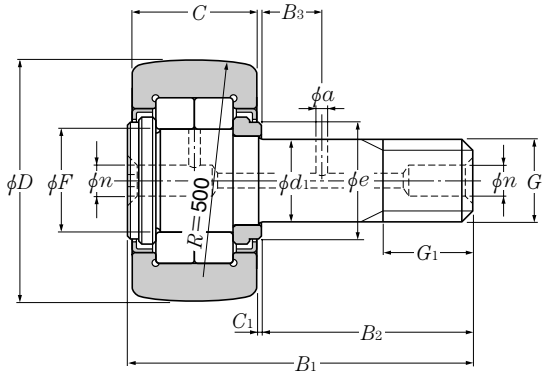
Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
16～26	NIP-X30	M4×0.7 ×4ℓ	1M 6×1 ～1M10×1.25
30～35	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5～1M16×1.5
40～90	JIS A-PT1/8	PT1/8×7ℓ	1M18×1.5～1M30×1.5

※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Load ratings of truck (N)		Load ratings of truck (kgf)		Limiting speeds (r/min)		Maximum tightening torque		Bearing numbers		Mass (approx.) kg	
Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	grease	oil	N·m	kgf·m	Type KRU [1] Type KRVU	Type KRU··LL [1] Type KRVU··LL	Type KRU	Type KRVU
10 400	56 000	1 060	5 750	4 000	5 500	360	37	**KRU85**	**KRU85LL**	1.74	—
11 400	59 000	1 160	6 100	4 000	5 500	360	37	**KRU90**	**KRU90LL**	1.95	—
11 400	59 000	1 160	6 100	2 600	3 500	360	37	**KRVU90**	**KRVU90LL**	—	1.97

Metric Series

Type NUKR

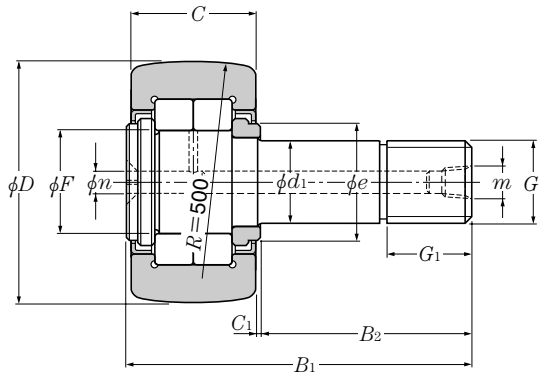


Type NUKR(D<**100mm**) (**Double row type full complement with seal**)

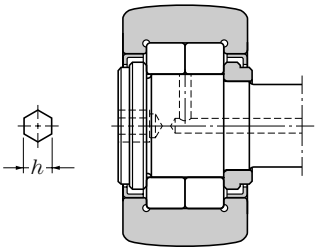
D 30～180mm

Boundary dimensions mm														
D $^{0}_{-0.050}$	C	d_1	F	B_1	B_2	G	G_1	C_1	n	m	B_3	a	e	h
30	14	12 $^{0}_{-0.018}$	14.5	40	25	M12×1.5	13	0.6	6	—	6	3	15	6
35	18	16 $^{0}_{-0.018}$	19	52	32.5	M16×1.5	17	0.8	6	—	8	3	21	6
40	20	18 $^{0}_{-0.018}$	21.5	58	36.5	M18×1.5	19	0.8	6	—	8	3	23	6
47	24	20 $^{0}_{-0.021}$	25.5	66	40.5	M20×1.5	21	0.8	8	—	9	4	27	8
52	24	20 $^{0}_{-0.021}$	30	66	40.5	M20×1.5	21	0.8	8	—	9	4	31	8
62	29	24 $^{0}_{-0.021}$	35	80	49.5	M24×1.5	25	0.8	8	—	11	4	38	8
72	29	24 $^{0}_{-0.021}$	41.5	80	49.5	M24×1.5	25	0.8	8	—	11	4	44	8
80	35	30 $^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	8	—	15	4	51	8
90	35	30 $^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	8	—	15	4	51	8
100	43	36 $^{0}_{-0.025}$	48.5	120	74	M36×1.5	38	1.5	8	PT$^{1}/_{8}$	—	—	53	14
120	50	42 $^{0}_{-0.025}$	60.5	140	87	M42×1.5	44	1.5	8	PT$^{1}/_{8}$	—	—	66	14
140	57	48 $^{0}_{-0.025}$	65	160	100	M48×1.5	52	1.5	8	PT$^{1}/_{8}$	—	—	72.5	14
150	60	52 $^{0}_{-0.030}$	75.5	170	107	M52×1.5	52	1.5	8	PT$^{1}/_{8}$	—	—	85.5	17
160	63	56 $^{0}_{-0.030}$	80.5	180	113	M56×3	58	2	8	PT$^{1}/_{8}$	—	—	89.5	17
170	66	60 $^{0}_{-0.030}$	86	190	120	M60×3	58	2	8	PT$^{1}/_{8}$	—	—	96.5	17
180	72	64 $^{0}_{-0.030}$	91.5	200	124	M64×3	65	2	8	PT$^{1}/_{8}$	—	—	103.5	17

Note 1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter D which conforms to JIS Class-0. **Ex. NUKR52X**



Type NUKR（D≧100mm）

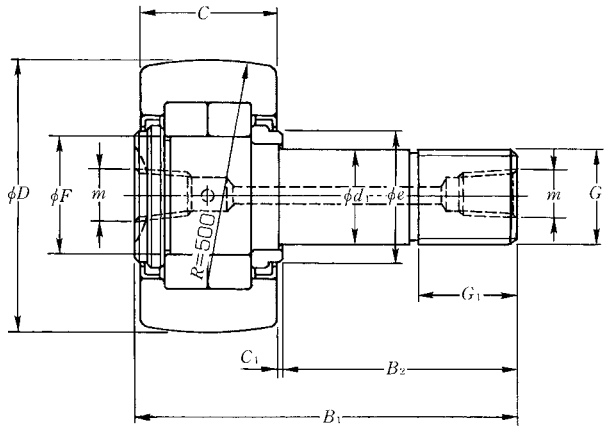


Type NUKR··H
(With hexagonal wrench socket)

Basic load ratings dynamic N C_r	Basic load ratings static N C_{or}	Basic load ratings dynamic kgf C_r	Basic load ratings static kgf C_{or}	Load ratings of truck N Spherical outer ring	Load ratings of truck N Cylindrical outer ring	Load ratings of truck kgf Spherical outer ring	Load ratings of truck kgf Cylindrical outer ring	Limiting speeds r/min grease	Maximum tightening torque N·m	Maximum tightening torque kgf·m	Bearing numbers Type KUKR[1)]	Mass (approx.) kg Type KUKR
13 300	13 500	1 360	1 380	2 620	7 700	267	785	6 900	20	2	**NUKR 30**	0.088
22 300	25 700	2 280	2 620	3 200	11 900	325	1 220	5 500	52	5.3	**NUKR 35**	0.165
24 100	29 100	2 450	2 970	3 850	14 500	390	1 480	4 700	76	7.8	**NUKR 40**	0.242
38 500	48 000	3 950	4 900	4 700	21 000	480	2 150	4 000	98	10	**NUKR 47**	0.380
42 500	57 500	4 350	5 850	5 550	23 300	565	2 370	3 300	98	10	**NUKR 52**	0.450
56 500	72 500	5 750	7 400	6 950	34 500	710	3 500	2 900	178	18	**NUKR 62**	0.795
62 000	85 500	6 350	8 700	8 050	38 500	820	3 900	2 400	178	18	**NUKR 72**	1.01
101 000	151 000	10 300	15 400	9 800	53 000	1 000	5 400	2 100	360	37	**NUKR 80**	1.54
101 000	151 000	10 300	15 400	11 400	59 000	1 160	6 100	2 100	360	37	**NUKR 90**	1.96
119 000	167 000	12 100	17 000	13 000	79 000	1 300	8 050	2 000	630	65	**NUKR100**	3.08
172 000	266 000	17 600	27 100	16 400	113 000	1 670	11 500	1 700	1 020	105	**NUKR120**	5.17
201 000	294 000	20 500	30 000	20 000	152 000	2 040	15 500	1 500	1 540	160	**NUKR140**	7.98
258 000	380 000	26 300	39 000	22 000	173 000	2 250	17 600	1 300	1 950	200	**NUKR150**	9.70
274 000	400 000	27 900	41 000	24 000	194 000	2 450	19 800	1 200	2 480	250	**NUKR160**	11.7
320 000	475 000	32 500	48 500	26 000	218 000	2 650	22 200	1 100	3 030	310	**NUKR170**	13.9
365 000	555 000	37 500	56 500	27 900	253 000	2 840	25 800	1 000	3 670	375	**NUKR180**	17.0

Metric Series

Type NUKRT
Type NUKRT··X◦



Type NUKRT
(Shielded full-complement roller, double -row cylindrical roller type)

D 30～180mm

Boundary dimensions mm											Basic load ratings dynamic static dynamic static N kgf			
D [1)]	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	C_r	C_{or}	C_r	C_{or}
30	14	12 $^{0}_{-0.018}$	14.5	40	25	M12×1.5	13	0.6	15	M6×0.75	13 300	13 500	1 360	1 380
35	18	16 $^{0}_{-0.018}$	19	52	32.5	M16×1.5	17	0.8	21	PT1/8	22 300	25 700	2 280	2 620
40	20	18 $^{0}_{-0.018}$	21.5	58	36.5	M18×1.5	19	0.8	23	PT1/8	24 100	29 100	2 450	2 970
47	24	20 $^{0}_{-0.021}$	25.5	66	40.5	M20×1.5	21	0.8	27	PT1/8	38 500	48 000	3 950	4 900
52	24	20 $^{0}_{-0.021}$	30	66	40.5	M20×1.5	21	0.8	31	PT1/8	42 500	57 500	4 350	5 850
62	29	24 $^{0}_{-0.021}$	35	80	49.5	M24×1.5	25	0.8	38	PT1/8	56 500	72 500	5 750	7 400
72	29	24 $^{0}_{-0.021}$	41.5	80	49.5	M24×1.5	25	0.8	44	PT1/8	62 000	85 500	6 350	8 700
80	35	30 $^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	51	PT1/8	101 000	151 000	10 300	15 400
90	35	30 $^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	51	PT1/8	101 000	151 000	10 300	15 400
100	43	36 $^{0}_{-0.025}$	48.5	120	74	M36×1.5	38	1.5	53	PT1/8	119 000	167 000	12 100	17 000
120	50	42 $^{0}_{-0.025}$	60.5	140	87	M42×1.5	44	1.5	66	PT1/8	172 000	266 000	17 600	27 100
140	57	48 $^{0}_{-0.025}$	65	160	100	M48×1.5	52	1.5	72.5	PT1/8	201 000	294 000	20 500	30 000
150	60	52 $^{0}_{-0.030}$	75.5	170	107	M52×1.5	52	1.5	85.5	PT1/8	258 000	380 000	26 300	39 000
160	63	56 $^{0}_{-0.030}$	80.5	180	113	M56×3	58	2	89.5	PT1/8	274 000	400 000	27 900	41 000
170	66	60 $^{0}_{-0.030}$	86	190	120	M60×3	58	2	96.5	PT1/8	320 000	475 000	32 500	48 500
180	72	64 $^{0}_{-0.030}$	91.5	200	124	M64×3	65	2	103.5	PT1/8	365 000	555 000	37 500	56 500

Note 1) The dimensional tolerance for the outer ring outer diameter D of **Type NUKRT··X** with cylindrical outer surface conforms to JIS Class-0.

Accessories

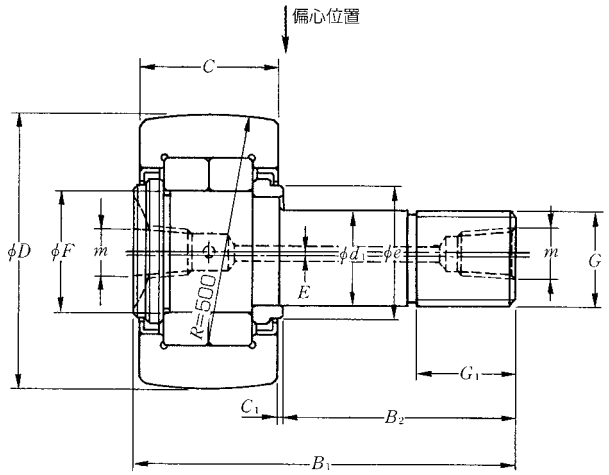
Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
30	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5
35～180	JIS A-PT1/8	PT1/8×7 ℓ	1M16×1.5～1M64×3

※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Load ratings of truck N		Load ratings of truck kgf		Limiting speeds r/min	Maximum tightening torque N·m	Maximum tightening torque kgf·m	Bearing numbers		Mass kg	Stud diameter mm
Spherical outer ring	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	grease			Cylindrical outer ring	Spherical outer ring	(approx.)	
2 620	7 700	267	785	6 900	20	2	**NUKRT30**	**NUKRT30X**	0.088	**12**
3 200	11 900	325	1 220	5 500	52	5.3	**NUKRT35**	**NUKRT35X**	0.165	**16**
3 850	14 500	390	1 480	4 700	76	7.8	**NUKRT40**	**NUKRT40X**	0.242	**18**
4 700	21 000	480	2 150	4 000	98	10	**NUKRT47**	**NUKRT47X**	0.380	**20**
5 550	23 300	565	2 370	3 300	98	10	**NUKRT52**	**NUKRT52X**	0.450	**20**
6 950	34 500	710	3 500	2 900	178	18	**NUKRT62**	**NUKRT62X**	0.795	**24**
8 050	38 500	820	3 900	2 400	178	18	**NUKRT72**	**NUKRT72X**	1.01	**24**
9 800	53 000	1 000	5 400	2 100	360	37	**NUKRT80**	**NUKRT80X**	1.54	**30**
11 400	59 000	1 160	6 100	2 100	360	37	**NUKRT90**	**NUKRT90X**	1.96	**30**
13 000	79 000	1 300	8 050	2 000	630	65	**NUKRT100**	**NUKRT100X**	3.08	**36**
16 400	113 000	1 670	11 500	1 700	1 020	105	**NUKRT120**	**NUKRT120X**	5.17	**42**
20 000	152 000	2 040	15 500	1 500	1 540	160	**NUKRT140**	**NUKRT140X**	7.98	**48**
22 000	173 000	2 250	17 600	1 300	1 950	200	**NUKRT150**	**NUKRT150X**	9.70	**52**
24 000	194 000	2 450	19 800	1 200	2 480	250	**NUKRT160**	**NUKRT160X**	11.7	**56**
26 000	218 000	2 650	22 200	1 100	3 030	310	**NUKRT170**	**NUKRT170X**	13.9	**60**
27 900	253 000	2 840	25 800	1 000	3 670	375	**NUKRT180**	**NUKRT180X**	17.0	**64**

Metric Series

Type NUKRU
Type NUKRU··X



Type NUKRU
(Shielded full-complement roller, double -row cylindrical roller type)

D 30～180mm

Boundary dimensions mm												Basic load ratings N	
$D^{1)}$	C	d_1	F	B_1	B_2	G	G_1	C_1	e	m	E eccentricity	dynamic C_r	static C_{or}
30	14	$12\ ^{0}_{-0.018}$	14.5	40	25	M12×1.5	13	0.6	15	M6×0.75	0.4	13 300	13 500
35	18	$16\ ^{0}_{-0.018}$	19	52	32.5	M16×1.5	17	0.8	21	M6×0.75	0.5	22 300	25 700
40	20	$18\ ^{0}_{-0.018}$	21.5	58	36.5	M18×1.5	19	0.8	23	PT1/8	0.6	24 100	29 100
47	24	$20\ ^{0}_{-0.021}$	25.5	66	40.5	M20×1.5	21	0.8	27	PT1/8	0.7	38 500	48 000
52	24	$20\ ^{0}_{-0.021}$	30	66	40.5	M20×1.5	21	0.8	31	PT1/8	0.7	42 500	57 500
62	29	$24\ ^{0}_{-0.021}$	35	80	49.5	M24×1.5	25	0.8	38	PT1/8	0.8	56 500	72 500
72	29	$24\ ^{0}_{-0.021}$	41.5	80	49.5	M24×1.5	25	0.8	44	PT1/8	1.0	62 000	85 500
80	35	$30\ ^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	51	PT1/8	1.0	101 000	151 000
90	35	$30\ ^{0}_{-0.021}$	47.5	100	63	M30×1.5	32	1	51	PT1/8	1.0	101 000	151 000
100	43	$36\ ^{0}_{-0.025}$	48.5	120	74	M36×1.5	38	1.5	53	PT1/8	1.5	119 000	167 000
120	50	$42\ ^{0}_{-0.025}$	60.5	140	87	M42×1.5	44	1.5	66	PT1/8	1.5	172 000	266 000
140	57	$48\ ^{0}_{-0.025}$	65	160	100	M48×1.5	52	1.5	72.5	PT1/8	2	201 000	294 000
150	60	$52\ ^{0}_{-0.030}$	75.5	170	107	M52×1.5	52	1.5	85.5	PT1/8	2	258 000	380 000
160	63	$56\ ^{0}_{-0.030}$	80.5	180	113	M56×3	58	2	89.5	PT1/8	2	274 000	400 000
170	66	$60\ ^{0}_{-0.030}$	86	190	120	M60×3	58	2	96.5	PT1/8	2.5	320 000	475 000
180	72	$64\ ^{0}_{-0.030}$	91.5	200	124	M64×3	65	2	103.5	PT1/8	2.5	365 000	555 000

Note 1) The dimensional tolerance for the outer ring outer diameter *D* of **Type NUKRU··X** with cylindrical outer surface conforms to JIS Class-0.

Accessories

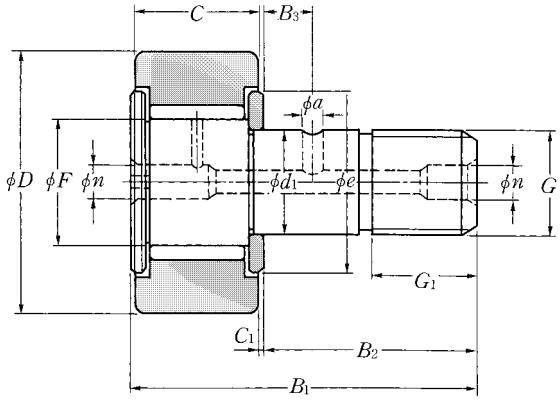
Nominal number of applied bearing	Nominal No. of grease nipple	Nominal No. of threaded plug with hexagon socket	Nominal No. of applied hexagon nut
30～ 35	JIS A-M6F	M6×0.75×6ℓ	1M12×1.5～1M16×1.5
40～180	JIS A-PT1/8	PT1/8×7 ℓ	1M18×1.5～1M64×3

※Table 3 on B-179 and Table 5-2 on B-180 shows the respective boundary dimensions of grease nipple and threaded plug with hexagon socket.

Basic load ratings dynamic kgf C_r	Basic load ratings static kgf C_{or}	Load ratings of truck N Spherical outer ring	Load ratings of truck N Cylindrical outer ring	Load ratings of truck kgf Spherical outer ring	Load ratings of truck kgf Cylindrical outer ring	Limiting speeds r/min grease	Maximum tightening torque N·m	Maximum tightening torque kgf·m	Bearing numbers Cylindrical outer ring	Bearing numbers Spherical outer ring	Mass kg (approx.)	Stud diameter mm
1 360	1 380	2 620	7 700	267	785	6 900	20	2	**NUKRU30**	**NUKRU30X**	0.088	**12**
2 280	2 620	3 200	11 900	325	1 220	5 500	52	5.3	**NUKRU35**	**NUKRU35X**	0.165	**16**
2 450	2 970	3 850	14 500	390	1 480	4 700	76	7.8	**NUKRU40**	**NUKRU40X**	0.242	**18**
3 950	4 900	4 700	21 000	480	2 150	4 000	98	10	**NUKRU47**	**NUKRU47X**	0.380	**20**
4 350	5 850	5 550	23 300	565	2 370	3 300	98	10	**NUKRU52**	**NUKRU52X**	0.450	**20**
5 750	7 400	6 950	34 500	710	3 500	2 900	178	18	**NUKRU62**	**NUKRU62X**	0.795	**24**
6 350	8 700	8 050	38 500	820	3 900	2 400	178	18	**NUKRU72**	**NUKRU72X**	1.01	**24**
10 300	15 400	9 800	53 000	1 000	5 400	2 100	360	37	**NUKRU80**	**NUKRU80X**	1.54	**30**
10 300	15 400	11 400	59 000	1 160	6 100	2 100	360	37	**NUKRU90**	**NUKRU90X**	1.96	**30**
12 100	17 000	13 000	79 000	1 300	8 050	2 000	630	65	**NUKRU100**	**NUKRU100X**	3.08	**36**
17 600	27 100	16 400	113 000	1 670	11 500	1 700	1 020	105	**NUKRU120**	**NUKRU120X**	5.17	**42**
20 500	30 000	20 000	152 000	2 040	15 500	1 500	1 540	160	**NUKRU140**	**NUKRU140X**	7.98	**48**
26 300	39 000	22 000	173 000	2 250	17 600	1 300	1 950	200	**NUKRU150**	**NUKRU150X**	9.70	**52**
27 900	41 000	24 000	194 000	2 450	19 800	1 200	2 480	250	**NUKRU160**	**NUKRU160X**	11.7	**56**
32 500	48 500	26 000	218 000	2 650	22 200	1 100	3 030	310	**NUKRU170**	**NUKRU170X**	13.9	**60**
37 500	56 500	27 900	253 000	2 840	25 800	1 000	3 670	375	**NUKRU180**	**NUKRU180X**	17.0	**64**

Inch series

Type CRV··X◦
Type CRV··XLL



Type CR
(with cage)

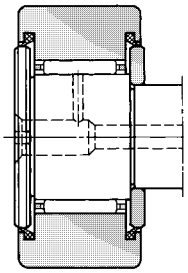
D 12.700～152.400mm

Boundary dimensions

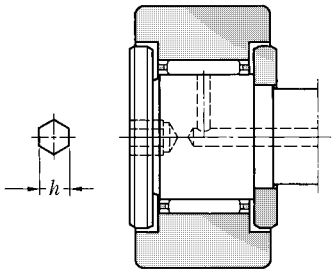
mm (1/25.4mm)

D $^{0}_{-0.025}$	*C* $^{0}_{-0.130}$	*d*1 $^{+0.025}_{0}$	*F*	*B*1	*B*2	*G*	*G*1	*C*1	*n*	*B*3	*a*	*e*	*h*
12.700(1/2)	8.731(11/32)	4.762(−)	6	23	12.7	NO-10-32UNF	6.4	0.8	3[2)]	—	—	10	1/8
	9.525(3/8)	4.762(−)	6	27	15.9	NO-10-32UNF	6.4	0.8	3[2)]	—	—	10	1/8
15.875(5/8)	10.319(13/32)	6.350(1/4)	8	27.8	15.9	1/4-28UNF	7.9	0.8	4[2)]	—	—	12.5	1/8
	11.112(7/16)	6.350(1/4)	8	31.8	19.1	1/4-28UNF	7.9	0.8	4[2)]	—	—	12.5	1/8
19.050(3/4)	12.700(1/2)	9.525(3/8)	12	36.5	22.2	3/8-24UNF	9.5	0.8	4	6.35	3	16.2	3/16
22.225(7/8)	12.700(1/2)	9.525(3/8)	12	36.5	22.2	3/8-24UNF	9.5	0.8	4	6.35	3	16.2	3/16
25.400(1)	15.875(5/8)	11.112(7/16)	13	42.8	25.4	7/16-20UNF	12.7	0.8	4	6.35	3	21	1/4
28.575(1 1/8)	15.875(5/8)	11.112(7/16)	13	42.8	25.4	7/16-20UNF	12.7	0.8	4	6.35	3	21	1/4
31.750(1 1/4)	19.050(3/4)	12.700(1/2)	16	52.4	31.8	1/2-20UNF	15.9	0.8	6	7.94	3	25	1/4
34.925(1 3/8)	19.050(3/4)	12.700(1/2)	16	52.4	31.8	1/2-20UNF	15.9	0.8	6	7.94	3	25	1/4
38.100(1 1/2)	22.225(7/8)	15.875(5/8)	20	61.9	38.1	5/8-18UNF	19.1	0.8	6	9.53	4	30	5/16
41.275(1 5/8)	22.225(7/8)	15.875(5/8)	20	61.9	38.1	5/8-18UNF	19.1	0.8	6	9.53	4	30	5/16
44.450(1 3/4)	25.400(1)	19.050(3/4)	25	71.4	44.4	3/4-16UNF	22.2	0.8	6	11.11	4	36.5	5/16
47.625(1 7/8)	25.400(1)	19.050(3/4)	25	71.4	44.4	3/4-16UNF	22.2	0.8	6	11.11	4	36.5	5/16
50.800(2)	31.750(1 1/4)	22.225(7/8)	30	84.1	50.8	7/8-14UNF	25.4	0.8	8	12.7	5	41.5	7/16
57.150(2 1/4)	31.750(1 1/4)	22.225(7/8)	30	84.1	50.8	7/8-14UNF	25.4	0.8	8	12.7	5	41.5	7/16

1) For bearing with cylindrical outer surface, the nominal number is followed by "**X**" code. In this case, the cylindrical bearing is manufactured with the tolerance for outer ring outer diameter *D* which conforms to JIS Class-0. **Ex. CR10XLL**



Type CR··LL
(with cage, sealed type)



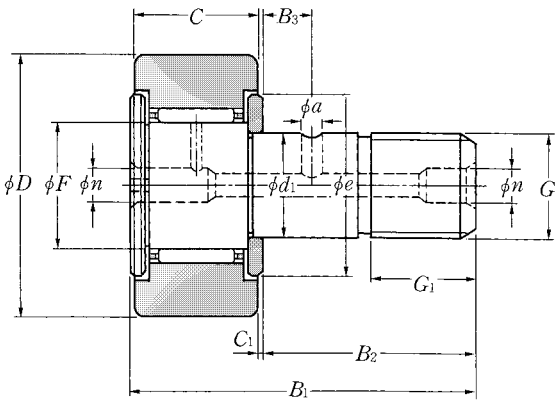
Type CR··H
(with hexagon hole)

Basic load ratings dynamic N C_r	Basic load ratings static N C_{or}	Basic load ratings dynamic kgf C_r	Basic load ratings static kgf C_{or}	Load ratings of truck N Spherical outer ring	Load ratings of truck N Cylindrical outer ring	Load ratings of truck kgf Spherical outer ring	Load ratings of truck kgf Cylindrical outer ring	Limiting speeds r/min grease	Limiting speeds r/min oil	Maximum tightening torque N·m	Maximum tightening torque kgf·m	Bearing numbers Type CRV··X[1)]	Bearing numbers Type CRV··XLL[1)]	Mass (approx.) kg Type CRV··X
2 820	2 450	287	250	2 090	790	213	81	※20 000	※28 000	2	0.2	CR8T2	———	0.009
2 820	2 450	287	250	2 310	790	235	81	※20 000	※28 000	2	0.2	CR8-1T2	———	0.010
4 050	4 000	415	405	3 000	1 080	310	110	※18 000	※25 000	4	0.4	CR10	———	0.020
4 050	4 000	415	405	3 300	1 080	335	110	※18 000	※25 000	4	0.4	CR10-1	CR10-1LL	0.022
5 300	6 650	540	680	4 600	1 380	470	140	13 000	※16 000	13	1.3	CR12	CR12LL	0.037
5 300	6 650	540	680	5 350	1 710	545	174	13 000	※16 000	13	1.3	CR14	CR14LL	0.048
7 250	8 350	740	850	7 400	2 060	755	210	12 000	15 000	18	1.9	CR16	CR16LL	0.087
7 250	8 350	740	850	8 350	2 430	850	248	12 000	15 000	18	1.9	CR18	CR18LL	0.100
11 400	15 900	1 160	1 620	11 400	2 840	1 160	290	9 000	13 000	24	2.4	CR20	CR20LL	0.150
11 400	15 900	1 160	1 620	12 500	3 250	1 280	330	9 000	13 000	24	2.4	CR22	CR22LL	0.166
13 300	20 100	1 360	2 120	16 300	3 600	1 660	365	7 500	10 000	51	5.2	CR24	CR24LL	0.235
13 300	20 100	1 360	2 120	17 600	4 050	1 800	410	7 500	10 000	51	5.2	CR26	CR26LL	0.265
20 700	33 500	2 110	3 450	21 600	4 400	2 200	450	6 000	8 000	92	9.3	CR28	CR28LL	0.375
20 700	33 500	2 110	3 450	23 200	4 850	2 360	495	6 000	8 000	92	9..3	CR30	CR30LL	0.420
28 900	55 000	2 950	5 600	31 000	5 300	3 150	540	5 000	6 600	150	15	CR32	CR32LL	0.505
28 900	55 000	2 950	5 600	35 000	6 200	3 550	635	5 000	6 600	150	15	CR36	CR36LL	0.750

Note 2) Grease replenishing hole is provided at front side only (on left side-face in the above sketch). However, in the case with "hexagon hole" no grease replenishing hole is provided.
Remarks: For the cam follower with ※marked seal, the allowable running speed is approximately 10 000 r/min.

Inch series

Type CRV··X◦
Type CRV··XLL



Type CRV··X
(Full-complement roller type)

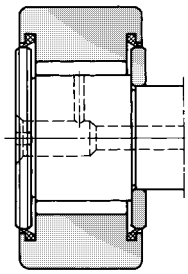
D 12.700～152.400mm

Boundary dimensions

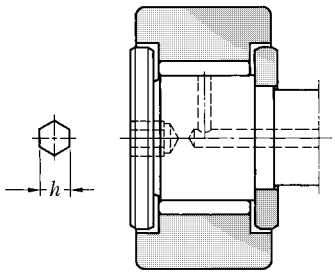
mm (1/25.4mm)

D $^{0}_{-0.025}$	*C* $^{0}_{-0.130}$	d_1 $^{+0.025}_{0}$	*F*	B_1	B_2	*G*	G_1	C_1	*n*	B_3	*a*	*e*	*h*
12.700(1/2)	8.731(11/32)	4.826(−)	5.75	23	12.7	NO-10-32UNF	6.4	0.8	3[2)]	—	—	10	1/8
	9.525(3/8)	4.826(−)	5.75	27	15.9	NO-10-32UNF	6.4	0.8	3[2)]	—	—	10	1/8
15.875(5/8)	10.319(13/32)	6.350(1/4)	8.11	27.8	15.9	1/4-28UNF	7.9	0.8	4[2)]	—	—	12.5	1/8
	11.112(7/16)	6.350(1/4)	8.11	31.8	19.1	1/4-28UNF	7.9	0.8	4[2)]	—	—	12.5	1/8
19.050(3/4)	12.700(1/2)	9.525(3/8)	11	36.5	22.2	3/8-24UNF	9.5	0.8	4	6.35	3	15.5	3/16
22.225(7/8)	12.700(1/2)	9.525(3/8)	11	36.5	22.2	3/8-24UNF	9.5	0.8	4	6.35	3	15.5	3/16
25.400(1)	15.875(5/8)	11.112(7/16)	14	43	25.4	7/16-20UNF	12.7	0.8	4	6.35	3	19.5	1/4
28.575(1 1/8)	15.875(5/8)	11.112(7/16)	14	43	25.4	7/16-20UNF	12.7	0.8	4	6.35	3	19.5	1/4
31.750(1 1/4)	19.050(3/4)	12.700(1/2)	18.47	52.4	31.8	1/2-20UNF	15.9	0.8	6	7.94	3	25	1/4
34.925(1 3/8)	19.050(3/4)	12.700(1/2)	18.47	52.4	31.8	1/2-20UNF	15.9	0.8	6	7.94	3	25	1/4
38.100(1 1/2)	22.225(7/8)	15.875(5/8)	21	61.9	38.1	5/8-18UNF	19.1	0.8	6	9.53	4	27	5/16
41.275(1 5/8)	22.225(7/8)	15.875(5/8)	21	61.9	38.1	5/8-18UNF	19.1	0.8	6	9.53	4	27	5/16
44.450(1 3/4)	25.400(1)	19.050(3/4)	24.65	71.4	44.5	3/4-16UNF	22.2	0.8	6	11.11	4	36.5	5/16
47.625(1 7/8)	25.400(1)	19.050(3/4)	24.65	71.4	44.5	3/4-16UNF	22.2	0.8	6	11.11	4	36.5	5/16
50.800(2)	31.750(1 1/4)	22.225(7/8)	26.71	84.1	50.8	7/8-14UNF	25.4	0.8	6	12.7	5	36.5	7/16
57.150(2 1/4)	31.750(1 1/4)	22.225(7/8)	26.71	84.1	50.8	7/8-14UNF	25.4	0.8	6	12.7	5	36.5	7/16
63.500(2 1/2)	38.100(1 1/2)	25.400(1)	31.15	96.8	57.2	1-14UNF	28.6	0.8	6	14.29	5	44	1/2
69.850(2 3/4)	38.100(1 1/2)	25.400(1)	31.15	96.8	57.2	1-14UNF	28.6	0.8	6	14.29	5	44	1/2
76.200(3)	44.450(1 3/4)	31.750(1 1/4)	36.85	109.5	63.5	1 1/4-12UNF	31.8	0.8	8	15.88	5	53	5/8
82.550(3 1/4)	44.450(1 3/4)	31.750(1 1/4)	36.85	109.5	63.5	1 1/4-12UNF	31.8	0.8	8	15.88	5	53	5/8
88.900(3 1/2)	50.800(2)	34.925(1 3/8)	44.5	122.2	69.9	1 3/8-12UNF	34.9	0.8	8	17.46	5	60	5/8
101.600(4)	57.150(2 1/4)	38.100(1 1/2)	44.5	147.6	88.9	1 1/2-12UNF	38.1	0.8	8	19.05	5	63	3/4
127.000(5)	69.850(2 3/4)	50.800(2)	68.7	200	128.6	2-12UNF	65.1	0.8	8	22.23	5	89	7/8
152.400(6)	82.550(3 1/4)	63.500(2 1/2)	81.35	236.6	152.4	2 1/2-12UNF	76.2	0.8	8	25.4	5	110	1

Note 1) For bearing with spherical outer surface, the nominal number is followed by "**X**" code. In this case, the spherical bearing (R=500) is manufactured with the tolerance 0,-0.05 for the outer ring outer diameter D.



Type CRV··XLL
(Full-complement roller and sealed type)

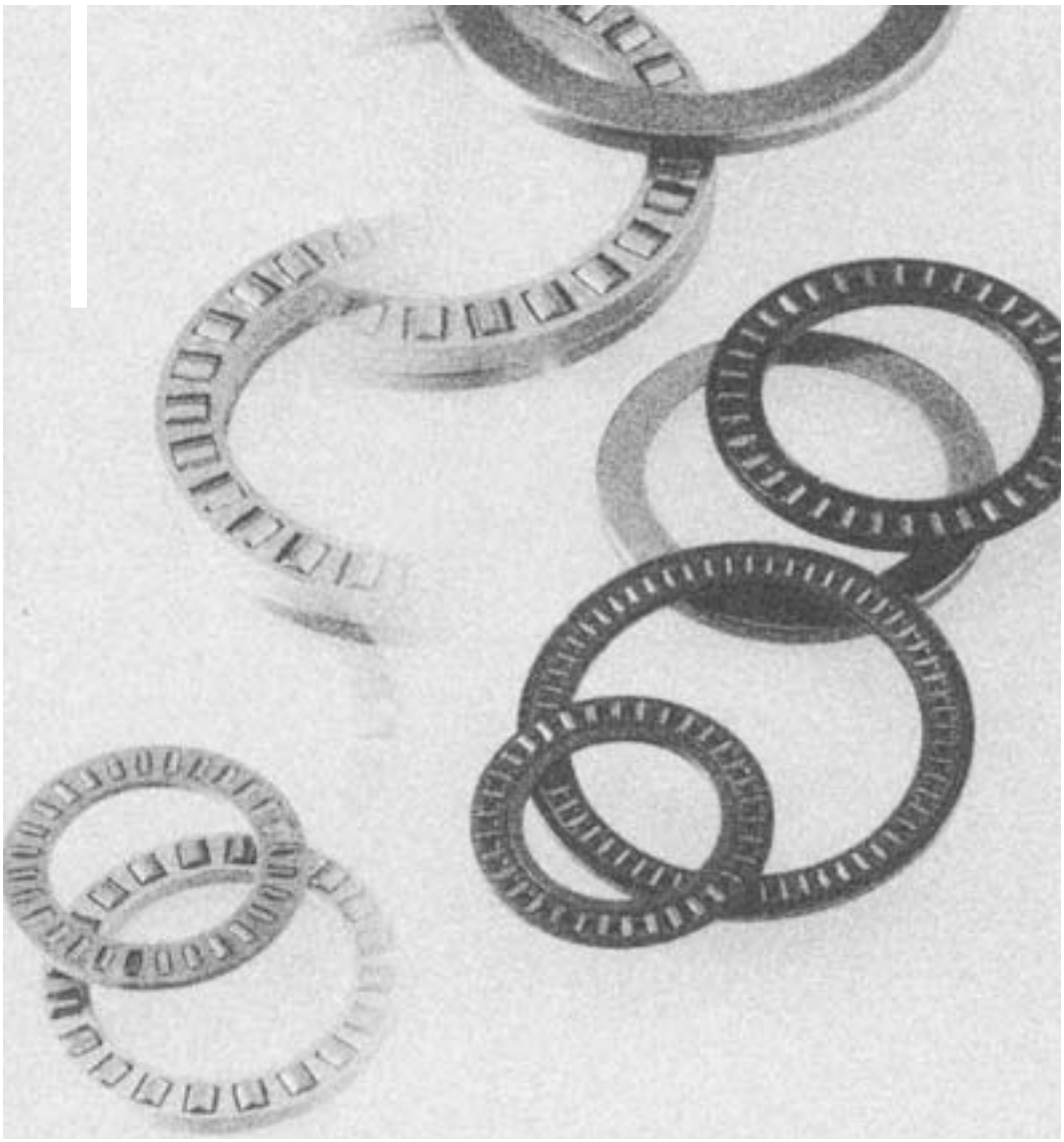


Type CRV··H
(with hexagon hole)

Basic load ratings				Load ratings of truck				Limiting speeds		Maximum tightening torque		Bearing numbers		Mass (approx.)
dynamic	static	dynamic	static	N		kgf		r/min						kg
	N		kgf	Spherical outer ring	Cylindrical outer ring	Spherical outer ring	Cylindrical outer ring	grease	oil	N·m	kgf·m	Type CRV··X[1)]	Type CRV··XLL[1)]	Type CRV··X
C_r	C_{or}	C_r	C_{or}											
3 400	3 750	350	385	2 090	790	213	81	※17 000	※22 000	2	0.2	CRV8X	CRV8XLL	0.010
3 950	4 550	405	465	2 310	790	235	81	※17 000	※22 000	2	0.2	CRV8-1X	CRV8-1XLL	0.011
5 550	7 600	565	770	3 000	1 080	310	110	※12 000	※15 000	4	0.4	CRV10X	CRV10XLL	0.020
6 200	8 700	630	885	3 300	1 080	335	110	※12 000	※15 000	4	0.4	CRV10-1X	CRV10-1XLL	0.022
8 050	13 300	825	1 360	4 600	1 380	470	140	9 000	※11 000	13	1.3	CRV12X	CRV12XLL	0.038
8 050	13 300	825	1 360	5 350	1 710	545	174	9 000	※11 000	13	1.3	CRV14X	CRV14XLL	0.048
11 700	18 900	1 190	1 920	7 400	2 060	755	210	7 100	9 200	18	1.9	CRV16X	CRV16XLL	0.080
11 700	18 900	1 190	1 920	8 350	2 430	850	248	7 100	9 200	18	1.9	CRV18X	CRV18XLL	0.096
17 700	35 000	1 810	3 600	11 400	2 840	1 160	290	5 400	7 000	24	2.4	CRV20X	CRV20XLL	0.140
17 700	35 000	1 810	3 600	12 500	3 250	1 280	330	5 400	7 000	24	2.4	CRV22X	CRV22XLL	0.165
21 100	45 500	2 150	4 650	16 300	3 600	1 660	365	4 800	6 200	51	5.2	CRV24X	CRV24XLL	0.240
21 100	45 500	2 150	4 650	17 600	4 050	1 800	410	4 800	6 200	51	5.2	CRV26X	CRV26XLL	0.280
28 400	60 500	2 900	6 150	21 600	4 400	2 200	450	4 100	5 300	92	9.3	CRV28X	CRV28XLL	0.400
28 400	60 500	2 900	6 150	23 200	4 850	2 360	495	4 100	5 300	92	9.3	CRV30X	CRV30XLL	0.440
41 000	87 500	4 200	8 950	31 000	5 300	3 150	540	3 700	4 800	150	15	CRV32X	CRV32XLL	0.650
41 000	87 500	4 200	8 950	35 000	6 200	3 550	635	3 700	4 800	150	15	CRV36X	CRV36XLL	0.780
54 500	119 000	5 600	12 200	44 500	7 200	4 550	735	3 200	4 100	230	23	CRV40X	CRV40XLL	1.20
54 500	119 000	5 600	12 200	49 000	8 250	5 000	840	3 200	4 100	230	23	CRV44X	CRV44XLL	1.34
76 500	177 000	7 800	18 000	64 000	9 150	6 500	935	2 700	3 500	435	45	CRV48X	CRV48XLL	1.92
76 500	177 000	7 800	18 000	69 000	10 000	7 050	1 020	2 700	3 500	435	45	CRV52X	CRV52XLL	2.20
84 500	214 000	8 650	21 800	86 500	11 100	8 800	1 130	2 200	2 800	580	60	CRV56X	CRV56XLL	2.92
106 000	244 000	10 800	24 900	113 000	13 200	11 500	1 350	2 200	2 800	760	78	CRV64X	CRV64XLL	4.32
189 000	520 000	19 300	53 000	165 000	17 900	16 900	1 830	1 500	1 900	1 820	190	CRV80X	CRV80XLL	8.80
260 000	675 000	26 500	68 500	240 000	22 100	24 400	2 250	1 200	1 500	3 550	360	CRV96X	CRV96XLL	15.3

Note 2) Grease replenishing hole is provided at front side only (on left side-face in the above sketch). However, in the case with "hexagon hole" no grease replenishing hole is provided.
Remarks: For the cam follower with ※marked seal, the allowable running speed is approximately 10 000 r/min.

Thrust Roller Bearings

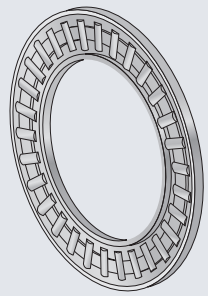
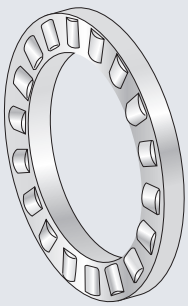
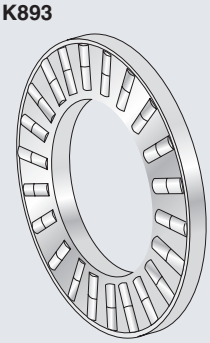


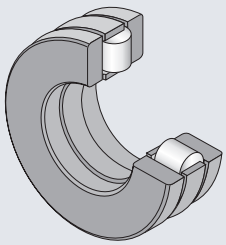
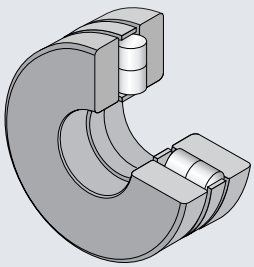
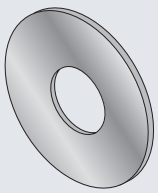
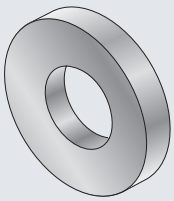
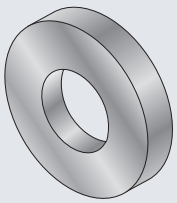
Thrust Roller Bearings

This thrust roller bearing composed of a thrust roller and cage assembly, wherein needle rollers or cylindrical rollers are configured radially in the cage, and a bearing ring of disc form is intended to support one-directional axial load.

In mounting, it is possible to use a shaft or a housing as the direct raceway surface without using the bearing ring, whereby design of a low height and lightweight compact construction is enabled.

This type of thrust roller bearing results in slipping on raceway surface because theoretically it can't roll perfectly, but in most cases it is practically trouble-free and can rotates at comparatively high speed.

Bearing type	Cage type	Applied shaft dia (mm)	Composition of nominal number	Nominal number	Remarks
AXK 	Punched steel plate cage	ϕ10～ϕ120	**AXK 11 04** 04: Bore diameter code 11: Dimension series code AXK: Type code	**AXK1104**	Possible to use in combination with AS type bearing ring.
	High tensile brass cage	ϕ130～ϕ160			
K811 K812 	Standard type Polyamide resin cage	Type K811 ϕ10～ϕ120 Type K812 ϕ30～ϕ80	**K8 11 10 T2** T2: Tail code 10: Bore diameter code 11: Dimension series code K8: Type code **[Tail code]** T2: resin cage JW: Punched steel plate cage	**K81110T2**	Any bearing type with tail code T2, that is, with polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. Feel free to contact **NTN** for the detail of the punched steel plate cage. Possible to use in combination with GS and WS bearing rings. K811 conforms to the Dimension Series 11 specified in JIS B 1512. K812 conforms to the Dimension Series 12 specified in JIS B 1512.
	Aluminum alloy cage	Type K811 ϕ130～ϕ160 Type K812 ϕ85～ϕ140			
	Punched steel plate cage	ϕ10～ϕ90			
K893 	Aluminum alloy cage	ϕ30～ϕ110	**K8 93 10** 10: Bore diameter code 93: Dimension series code K8: Type code	**K89310**	K893 conforms to the Dimension Series 93 specified in JIS B 1512.

Bearing type	Applied shaft dia (mm)	Composition of nominal number	Nominal number	Remarks
811 **812** 	ϕ10～ϕ160	**8 11 10 T2** Tail code Bore diameter code Dimension series code Type code **[Tail code]** T2: resin cage J: Punched steel plate cage	**81110T2**	Any bearing type with tail code T2, that is, with polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less. WS and GS bearing rings are used in set. 811 conforms to the Dimension Series 11 specified in JIS B 1512. 812 conforms to the Dimension Series 12 specified in JIS B 1512.
893 	ϕ30～ϕ110	**8 93 10** Bore diameter code Dimension series code Type code	**89310**	WS and GS bearing rings are used in set. 893 conforms to the Dimension Series 93 specified in JIS B 1512.
AS  Punched steel plate cage	ϕ10～ϕ130	**AS 11 04** Bore diameter code Dimension series code Type code	**AS1104**	Because of its 1mm thick steel plate ring, this thrust bearing needs adequate rigidity and profile accuracy of machine parts adjacent to the bearing. As an individual, on occasion this thrust bearing results in slight camber, but it is flattened when specific thrust load acts thereon, having no problem in practical use.
WS811 **WS812**  Solid type bore guide	ϕ10～ϕ160	**WS8 11 04** Bore diameter code Dimension series code Type code	**WS81104**	Higher rigidity and higher running accuracy than AS bearing ring .
GS811 **GS812**  Solid type outer guide	ϕ10～ϕ160	**GS8 11 04** Bore diameter code Dimension series code Type code	**GS81104**	Higher rigidity and higher running accuracy than AS bearing ring .

Bearing accuracy

The respective dimensional accuracy, profile accuracy and running accuracy of thrust cylindrical roller bearings 811, 812 and 893 are as shown in **Table 4.4** of Section 4 "**Bearing Accuracy**" (page A-28).

The thrust roller and cage assemblies (**AXK**, **K811**, **K812**, **K893**) are machined with dimensional tolerance E11 for the bore diameter (D_{e1}) (E12 for bearing number with T2 code) and with dimensional tolerance c12 for the outer diameter (D_e) of **Type AXK** and a13 for the diameter (D_e) of **K811**, **K812** and **K893**.

Raceway surface requirements:

Where the plane portion of a shaft/a housing is used as the direct raceway surface of thrust roller and cage assembly, the raceway surface must meet the requirements specified as a guideline in **Table 1**.

Table 1 Raceway surface requirements

Characteristics	Specified requirements
Perpendicularity (Max)	IT6 (IT4)
Surface roughness	0.2a
Surface hardness	HRC58～64
Effective hardened layer depth	Refer to Formula (7.1) on page A-39.

Reference : The parenthesized value shall be applied for high running accuracy.

Cage guiding

The thrust roller and cage assemblies (**AXK**, **K811**, **K812** and **K893**) must be guided at either the bore side (shaft side) or the outer surface side (housing side) for in-running centering.

In general, the bore-side guide of low relative speed against the cage is mostly used. It should be used particularly for high speed running. The dimensional tolerances for shaft and housing, when the cage is guided thereby, shall be h8 for shaft diameter (bore guide) and H9 for housing bore diameter (outer surface guide) respectively, which of the guide surface shall be fine-finished by grinding.

Bearing fit in bearing ring

Table 2 shows the tolerances for fitting of the thrust bearing rings (**AS**, **WS** and **GS**) on shaft or in housing.

Table 2 Bearing ring fit in shaft and housing

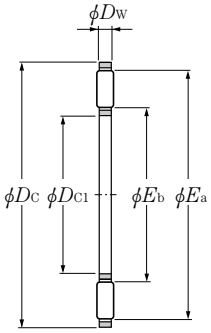
Bearing ring	Shaft	Housing
Series AS	h10	H11
Series WS	h6	—
Series GS	—	H7

Mounting related dimensions

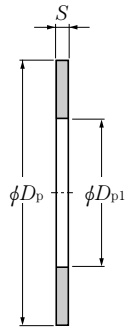
The dimensions related to fitting of **WS** and **GS** bearing rings on shaft and in housing are as described in applicable Dimensions Table.

The fitting surface of **AS** bearing ring must be flat and have the rigidity sufficient to support thrust load throughout its entire surface.

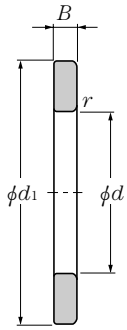
Type AXK11
Type AS11
Type WS811
Type GS811



Type AXK (Thrust needle roller and cage assy)



AS bearing ring (washer)

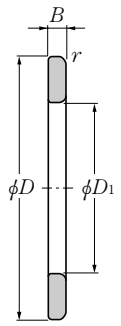


WS bearing ring (Inner ring)

D_{c1} 10～140mm

Boundary dimensions mm											Basic load ratings dynamic N	static N	dynamic kgf	static kgf
D_{c1} E11	D_c c12	D_w 0 -0.01	D_p e13	D_{p1} E12	S 2) 0.05	d	d_1 -0.2 -0.5	D	D_1 +0.5 +0.2	B	$r_{s\,min}$ 1)	C_a	C_{oa}	C_a	C_{oa}
10	24	2	24	10	1	10	24	24	10	$2.75^{0}_{-0.060}$	0.3	9 150	25 300	935	2 580
12	26	2	26	12	1	12	26	26	12	$2.75^{0}_{-0.060}$	0.3	9 850	28 900	1 010	2 940
15	28	2	28	15	1	15	28	28	16	$2.75^{0}_{-0.060}$	0.3	11 300	36 000	1 150	3 700
17	30	2	30	17	1	17	30	30	18	$2.75^{0}_{-0.060}$	0.3	11 900	39 500	1 220	4 050
20	35	2	35	20	1	20	35	35	21	$2.75^{0}_{-0.060}$	0.3	13 200	46 500	1 340	4 750
25	42	2	42	25	1	25	42	42	26	$3^{0}_{-0.060}$	0.6	14 600	58 000	1 490	5 900
30	47	2	47	30	1	30	47	47	32	$3^{0}_{-0.060}$	0.6	16 300	69 500	1 660	7 100
35	52	2	52	35	1	35	52	52	37	$3.5^{0}_{-0.075}$	0.6	17 800	81 500	1 820	8 300
40	60	3	60	40	1	40	60	60	42	$3.5^{0}_{-0.075}$	0.6	27 400	110 000	2 790	11 300
45	65	3	65	45	1	45	65	65	47	$4^{0}_{-0.075}$	0.6	29 800	128 000	3 050	13 100
50	70	3	70	50	1	50	70	70	52	$4^{0}_{-0.075}$	0.6	31 500	143 000	3 250	14 500
55	78	3	78	55	1	55	78	78	57	$5^{0}_{-0.075}$	0.6	38 000	186 000	3 850	19 000
60	85	3	85	60	1	60	85	85	62	$4.75^{0}_{-0.075}$	1	44 500	234 000	4 550	23 900
65	90	3	90	65	1	65	90	90	67	$5.25^{0}_{-0.075}$	1	46 500	254 000	4 750	25 900
70	95	4	95	70	1	70	95	95	72	$5.25^{0}_{-0.075}$	1	53 500	253 000	5 500	25 800
75	100	4	100	75	1	75	100	100	77	$5.75^{0}_{-0.075}$	1	55 000	266 000	5 650	27 100
80	105	4	105	80	1	80	105	105	82	$5.75^{0}_{-0.075}$	1	56 500	279 000	5 750	28 400
85	110	4	110	85	1	85	110	110	87	$5.75^{0}_{-0.075}$	1	57 500	291 000	5 900	29 700
90	120	4	120	90	1	90	120	120	92	$6.5^{0}_{-0.090}$	1	71 000	390 000	7 250	39 500
100	135	4	135	100	1	100	135	135	102	$7^{0}_{-0.090}$	1	90 500	550 000	9 200	56 500
110	145	4	145	110	1	110	145	145	112	$7^{0}_{-0.090}$	1	93 500	590 000	9 550	60 500
120	155	4	155	120	1	120	155	155	122	$7^{0}_{-0.090}$	1	99 000	650 000	10 100	66 500
130	170	5	170	130	1	130	170	170	132	$9^{0}_{-0.090}$	1	140 000	900 000	14 300	92 000
140	180	5	180	140	1	140	178	180	142	$9.5^{0}_{-0.090}$	1	145 000	960 000	14 800	97 500

Note 1) Allowable minimum chamfer dimension r.
2) Subject to measured thrust load of 20kg or more.



GS bearing ring (Outer ring)

Limiting speeds r/min		Bearing numbers				Reference dimensions mm		Mass kg (approx.)		
grease	oil	thrust needle roller and cage assembly	washer	inner ring	outer ring	E_b	E_a	AXK11	AS11	WS811 GS811
3 500	14 000	**AXK1100**	**AS1100**	**WS81100**	**GS81100**	12.3	21.7	0.0028	0.003	0.008
3 300	13 000	**AXK1101**	**AS1101**	**WS81101**	**GS81101**	14.3	23.7	0.003	0.0033	0.009
2 800	11 000	**AXK1102**	**AS1102**	**WS81102**	**GS81102**	17.2	26.5	0.0035	0.0035	0.01
2 500	10 000	**AXK1103**	**AS1103**	**WS81103**	**GS81103**	19.2	28.5	0.004	0.0038	0.011
2 100	8 500	**AXK1104**	**AS1104**	**WS81104**	**GS81104**	21.3	31.3	0.005	0.0051	0.014
1 800	7 000	**AXK1105**	**AS1105**	**WS81105**	**GS81105**	29.5	39.4	0.007	0.007	0.021
1 500	6 000	**AXK1106**	**AS1106**	**WS81106**	**GS81106**	34.5	44.4	0.008	0.008	0.025
1 400	5 500	**AXK1107**	**AS1107**	**WS81107**	**GS81107**	39.5	49.4	0.01	0.0091	0.033
1 200	4 700	**AXK1108**	**AS1108**	**WS81108**	**GS81108**	44.2	56.2	0.0185	0.0123	0.044
1 100	4 300	**AXK1109**	**AS1109**	**WS81109**	**GS81109**	50.5	62.4	0.0205	0.0136	0.055
1 000	3 900	**AXK1110**	**AS1110**	**WS81110**	**GS81110**	55.5	67.4	0.0235	0.0148	0.06
900	3 500	**AXK1111**	**AS1111**	**WS81111**	**GS81111**	61.0	74.9	0.0308	0.0189	0.095
800	3 200	**AXK1112**	**AS1112**	**WS81112**	**GS81112**	66.0	81.9	0.0390	0.0223	0.101
750	3 000	**AXK1113**	**AS1113**	**WS81113**	**GS81113**	71.0	86.9	0.04	0.0239	0.125
750	2 900	**AXK1114**	**AS1114**	**WS81114**	**GS81114**	75.5	91.4	0.06	0.0254	0.134
700	2 700	**AXK1115**	**AS1115**	**WS81115**	**GS81115**	80.5	96.4	0.061	0.027	0.155
650	2 600	**AXK1116**	**AS1116**	**WS81116**	**GS81116**	84.4	100.3	0.063	0.0284	0.163
600	2 400	**AXK1117**	**AS1117**	**WS81117**	**GS81117**	90.5	106.4	0.0668	0.0301	0.175
600	2 300	**AXK1118**	**AS1118**	**WS81118**	**GS81118**	96.5	116.4	0.086	0.0388	0.25
500	2 000	**AXK1120**	**AS1120**	**WS81120**	**GS81120**	107.5	131.4	0.112	0.0505	0.35
480	1 900	**AXK1122**	**AS1122**	**WS81122**	**GS81122**	115.5	139.4	0.122	0.0549	0.385
430	1 700	**AXK1124**	**AS1124**	**WS81124**	**GS81124**	125.5	149.4	0.131	0.0592	0.415
400	1 600	**AXK1126**	**AS1126**	**WS81126**	**GS81126**	136.0	164.0	0.205	0.074	0.663
380	1 500	**AXK1128**	**AS1128**	**WS81128**	**GS81128**	146.0	174.0	0.219	0.079	0.749